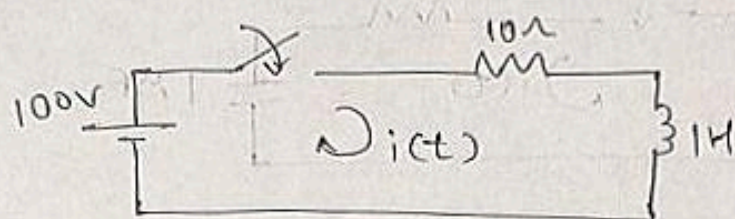


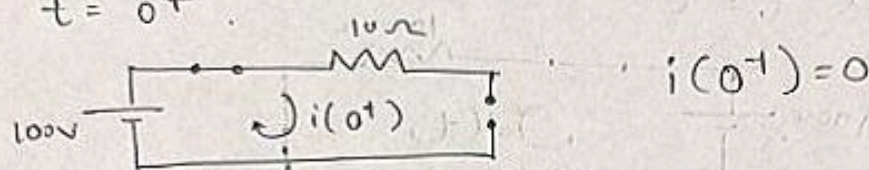
Module 4 - 01
 1. In the given circuit shown in fig the switch is closed at $t=0$. With zero current in inductor find i , $\frac{di}{dt}$ and $\frac{d^2i}{dt^2}$ at $t=0^+$



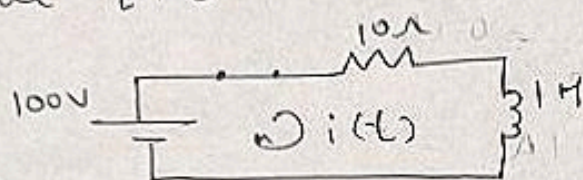
Solution:

→ at $t=0^-$, no current flow
 $i(0^-)=0$

→ at $t=0^+$



→ at $t > 0$



apply KVL

$$V(t) = V_R + V_L$$

$$V(t) = R i(t) + L \frac{di(t)}{dt}$$

$$100 = 10 i(t) + 1 \frac{di(t)}{dt}$$

$$10 i(t) + \frac{di(t)}{dt} - 100 = 0$$

$$\frac{di(t)}{dt} = 100 - 10 i(t) \quad \text{--- (1)}$$

at $t=0^+$

$$\frac{di(0^+)}{dt} = 100 - 10 i(0^+)$$

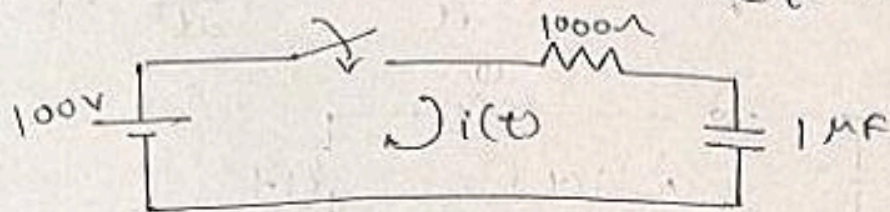
$$\boxed{\frac{di(0^+)}{dt} = 100 \text{ A/s}}$$

differentiate eq (1)

$$\frac{d^2i}{dt^2} = 0 - 10 \frac{di(t)}{dt} \Rightarrow 0 - 10(100)$$

$$\boxed{\frac{d^2i(t)}{dt^2} = -1000 \text{ A/s}^2}$$

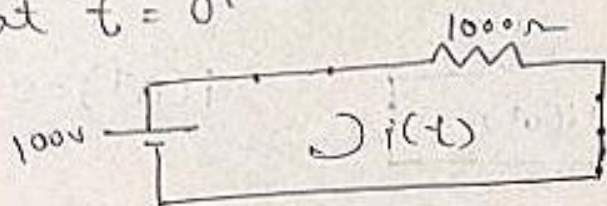
2. For n/w shown in figure, switch is closed at $t=0$. With capacitor uncharged find value for i , $\frac{di}{dt}$ & $\frac{d^2i}{dt^2}$ at $t=0^+$



solution

→ at $t=0^-$ $i(t)=0$
 $i(0^-)=0$

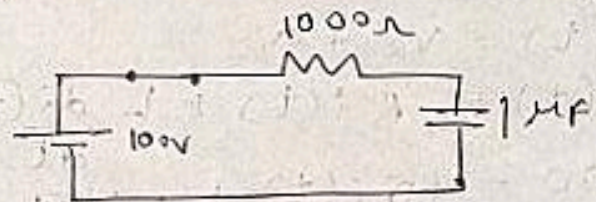
→ at $t=0^+$



$$i(t) = \frac{100}{1000} = 0.1 \text{ A}$$

@ $t=0^+$ $i(0^+) = 0.1 \text{ A}$
 $V_c(0^+) = 0$

→ at $t > 0$



Apply KVL

$$V_t = V_R + V_c$$

$$100 = R i(t) + \frac{1}{C} \int i(t) dt$$

$$100 = 1000 i(t) + \frac{1}{1 \times 10^{-6}} \int_0^t i(t) dt \quad \text{--- (1)}$$

$$100 - 1000 i(t) - \frac{1}{1 \times 10^{-6}} \int_0^t i(t) dt = 0$$

diff wrt t

$$0 - 1000 \frac{di(t)}{dt} - \frac{1}{1 \times 10^{-6}} i(t) = 0$$

$$-1000 \frac{di(t)}{dt} = \frac{1}{1 \times 10^{-6}} i(t)$$

$$\frac{di(t)}{dt} = -\frac{10^6}{10^3} i(t)$$

$$\frac{di(t)}{dt} = -1 \times 10^3 i(t) \rightarrow (3) \quad (2)$$

$$\text{at } t=0^+ \quad \frac{di(0^+)}{dt} = -1 \times 10^3 (0.1)$$

$$\boxed{\frac{di(0^+)}{dt} = -100 \text{ A/s}}$$

differentiate (3) w.r.t t .

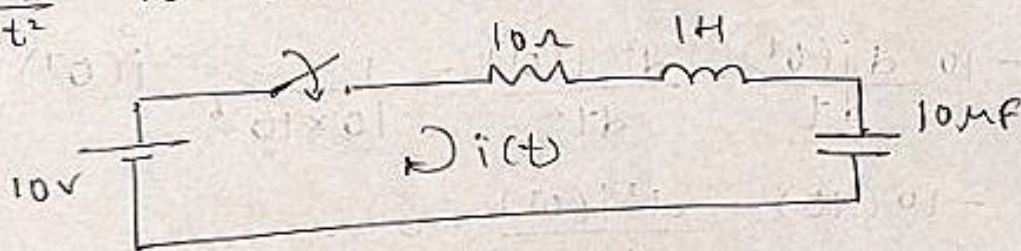
$$\frac{d^2 i(t)}{dt^2} = -10^3 \frac{di(t)}{dt}$$

$$\text{at } t=0^+, \quad \frac{d^2 i(0^+)}{dt^2} = -10^3 \frac{di(0^+)}{dt}$$

$$\frac{d^2 i(0^+)}{dt^2} = -1 \times 10^3 \times -100$$

$$\boxed{\frac{d^2 i(0^+)}{dt^2} = 100 \times 10^3 \text{ A/s}^2}$$

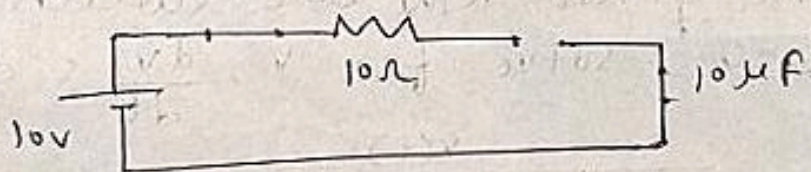
3. In a/cw shown switch is closed. Assuming initial conditions as 0 find i , $\frac{di}{dt}$ & $\frac{d^2 i}{dt^2}$ at $t=0^+$



solution

$$\rightarrow \text{at } t=0^- \quad \begin{aligned} i(t) &= 0 \\ i(0^-) &= 0 \\ V_c(0^-) &= 0 \end{aligned}$$

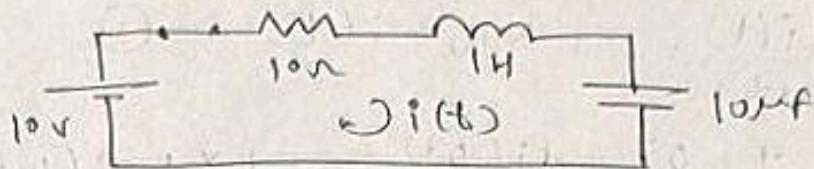
$$\rightarrow \text{at } t=0^+$$



$$i(0^+) = 0$$

$$V_c(0^+) = 0$$

→ at $t > 0$



apply KVL

$$v(t) = R i(t) + L \frac{di(t)}{dt} + \frac{1}{C} \int i(t) dt$$

$$10 = 10 i(t) + \frac{di(t)}{dt} + \frac{1}{10 \times 10^{-6}} \int i(t) dt$$

$$10 - 10 i(t) - \frac{di(t)}{dt} - \frac{1}{10 \times 10^{-6}} \int i(t) dt = 0 \quad \text{--- ①}$$

at $t = 0^+$

$$10 - 10 i(0^+) - \frac{di(0^+)}{dt} - \frac{1}{10 \times 10^{-6}} \int i(0^+) dt = 0$$

$$10 - \frac{di(0^+)}{dt} = 0$$

$$\Rightarrow \boxed{\frac{di(0^+)}{dt} = 10 \text{ A/s}}$$

diff ① wrt t

$$0 - 10 \frac{di(t)}{dt} - \frac{d^2 i(t)}{dt^2} - \frac{1}{10 \times 10^{-6}} i(t) = 0$$

at $t = 0^+$

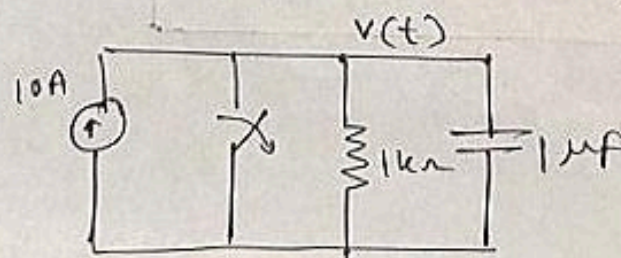
$$-10 \frac{di(0^+)}{dt} - \frac{d^2 i(0^+)}{dt^2} - \frac{1}{10 \times 10^{-6}} i(0^+) = 0$$

$$-10(10) - \frac{d^2 i(0^+)}{dt^2} = 0$$

$$\boxed{\frac{d^2 i(0^+)}{dt^2} = -100 \text{ A/s}^2}$$

4. In the given n/w the switch is opened at $t = 0$, solve for v , $\frac{dv}{dt}$ & $\frac{d^2 v}{dt^2}$ at $t = 0^+$

sol



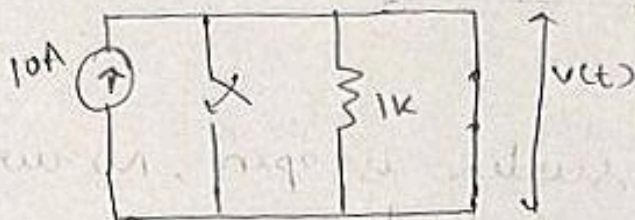
solution

→ at $t = 0^-$, switch is closed hence voltage across capacitor is zero

$$V_C(0^-) = 0$$

$$V(0^-) = 0$$

→ at $t = 0^+$

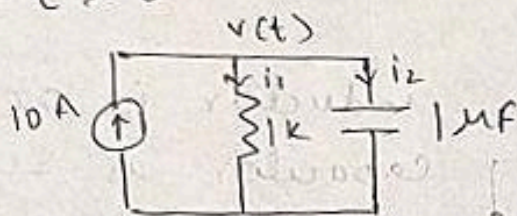


$$V_C(0^+) = 0$$

switch is open and capacitor behaves as short ckt at $t = 0^+$

$$\therefore V(0^+) = 0$$

→ at $t > 0$



Apply KCL, $i = i_1 + i_2$

$$\frac{v(t)}{1000} + 1 \times 10^{-6} \frac{dv(t)}{dt} = 10 \quad \text{--- (1)}$$

at $t = 0^+$

$$\frac{v(0^+)}{1000} + 1 \times 10^{-6} \frac{dv(0^+)}{dt} = 10$$

$$\frac{dv(0^+)}{dt} = \frac{10}{1 \times 10^{-6}} = 10 \times 10^6 \text{ V/s}$$

differentiate (1) w.r.t t

$$\frac{1}{1000} \frac{dv(t)}{dt} + 1 \times 10^{-6} \frac{d^2v(t)}{dt^2} = 0$$

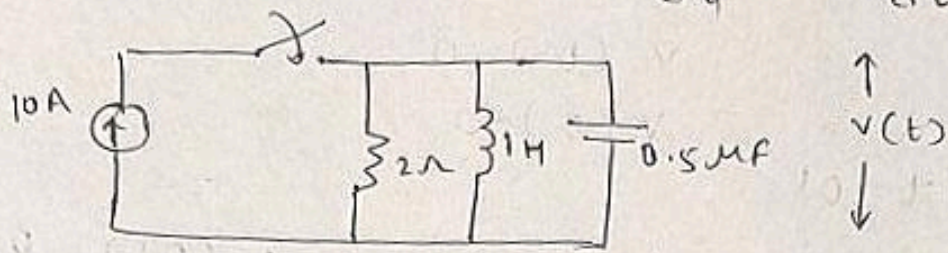
$$\frac{1}{1000} \frac{dv(t)}{dt} = -1 \times 10^{-6} \frac{d^2v(t)}{dt^2}$$

at $t = 0^+$

$$\frac{1}{1000} \frac{dv(0^+)}{dt} = -1 \times 10^{-6} \frac{d^2v(0^+)}{dt^2}$$

$$\frac{d^2v(0^+)}{dt^2} = \frac{-10 \times 10^6}{10^3 \times 10^{-6}} = -1 \times 10^{10} \text{ V/s}^2$$

5. For n/w shown in fig, switch is closed at $t=0$. determine V , $\frac{dV}{dt}$ & $\frac{d^2V}{dt^2}$ at $t=0^+$



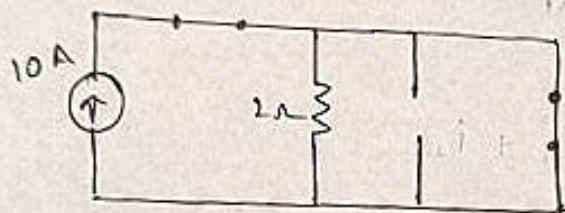
Solution

→ at $t=0^-$, switch is open, no current flows through inductor

$$i_L(0^-) = 0$$

$$V_C(0^-) = 0$$

→ at $t=0^+$

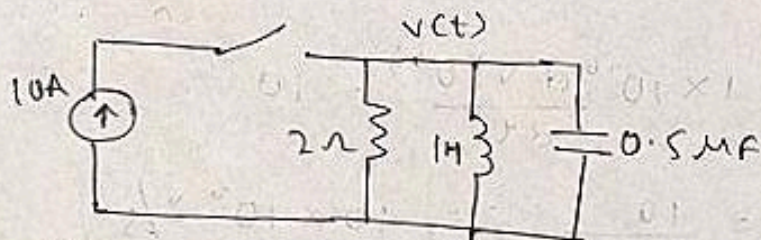


inductor is o.c.
capacitor is s.c.

$$i_L(0^+) = 0$$

$$V_C(0^+) = 0$$

→ for $t > 0$



apply KCL

$$10 = \frac{V(t)}{2} + \frac{1}{1} \int_0^t V(t) dt + 0.5 \times 10^{-6} \frac{dV(t)}{dt} \quad \text{--- (1)}$$

at $t=0^+$

$$\frac{V(0^+)}{2} + \frac{1}{1} \int_0^t V(0^+) dt + 0.5 \times 10^{-6} \frac{dV(0^+)}{dt} = 10$$

$$\frac{dV(0^+)}{dt} = \frac{10}{0.5 \times 10^{-6}} \quad \text{--- (2)}$$

$$\boxed{\frac{dV(0^+)}{dt} = 20 \times 10^6 \text{ V/s}}$$

differentiate (1) wrt t

(4)

$$\frac{1}{2} \frac{dv(t)}{dt} + v(t) + 0.5 \times 10^{-6} \frac{d^2 v(t)}{dt^2} = 0$$

at $t = 0^+$

$$\frac{1}{2} \frac{dv(0^+)}{dt} + v(0^+) + 0.5 \times 10^{-6} \frac{d^2 v(t)}{dt^2} = 0$$

$$0.5 \times 10^{-6} \frac{d^2 v(0^+)}{dt^2} = -0.5 \frac{dv}{dt}(0^+)$$

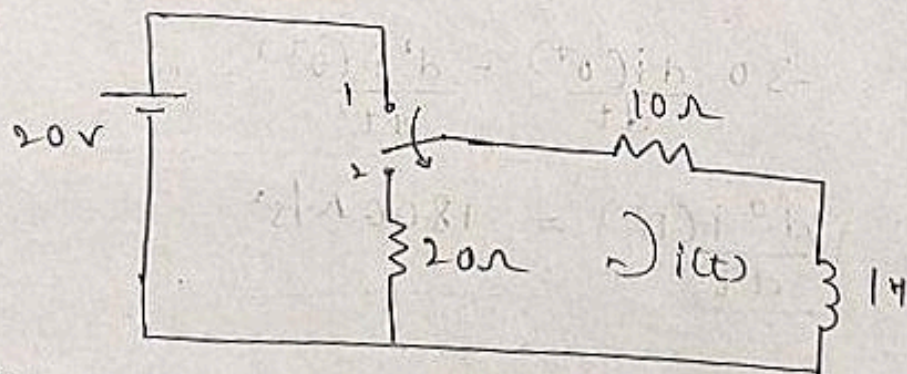
$$0.5 \times 10^{-6} \frac{d^2 v(0^+)}{dt^2} = -0.5 \times 20 \times 10^6$$

$$0.5 \times 10^{-6} \frac{d^2 v(0^+)}{dt^2} = -10 \times 10^6$$

$$\frac{d^2 v(0^+)}{dt^2} = \frac{-10 \times 10^6}{0.5 \times 10^{-6}}$$

$$\boxed{\frac{d^2 v(0^+)}{dt^2} = -20 \times 10^{12} \text{ V/s}^2}$$

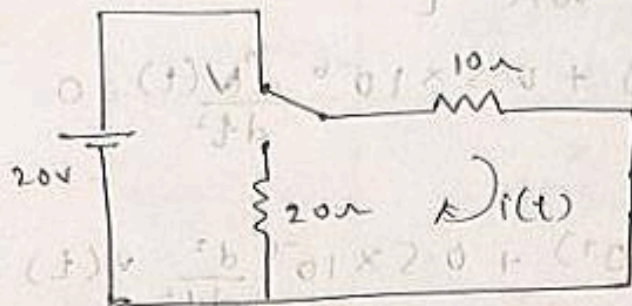
6. In the n/w shown, the switch is changed from position 1 to position 2 at $t = 0$, steady condition having reached before switching. Find values of i , $\frac{di}{dt}$ & $\frac{d^2 i}{dt^2}$ at $t = 0^+$



solution

→ at $t = 0^-$

since ckt is already in steady state inductor behaves as sc.



$$i(t) = \frac{20}{10 + 20} = 2 \text{ A}$$

$$i(0^-) = 2 \text{ A}$$

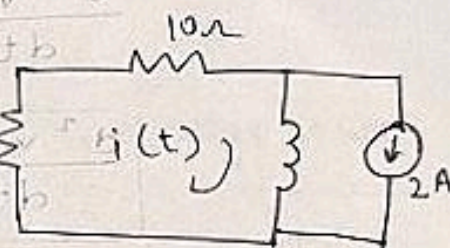
→ for $t > 0$, 10 ohm resistor is removed.

$$-20i(t) - 10 \frac{di(t)}{dt} = 0 \quad \text{--- (1)}$$

$$-30i(t) - \frac{di(t)}{dt} = 0$$

at $t = 0^+$, $i(0^+) = 2 \text{ A}$

$$-30i(0^+) - \frac{di(0^+)}{dt} = 0$$



differentiate (1)

$$-30 \frac{di(t)}{dt} - \frac{d^2 i(t)}{dt^2} = 0$$

at $t = 0^+$, $-30 \frac{di(0^+)}{dt} - \frac{d^2 i(0^+)}{dt^2} = 0$

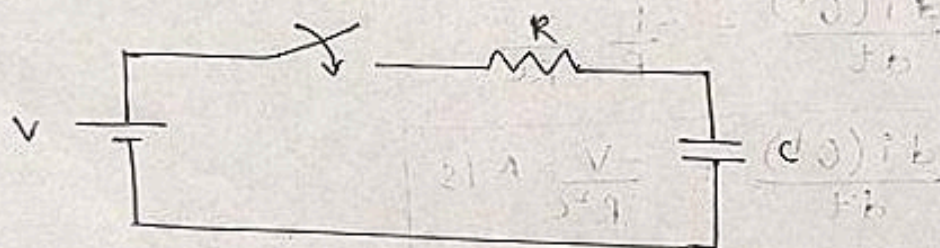
$$\frac{d^2 i(0^+)}{dt^2} = 1800 \text{ A/s}^2$$

7. Write behaviour of R, L, C elements at time of switching at $t=0$ both at $t=0^+$ & $t=\infty$

elements with initial condition	equivalent ckt at $t = 0^+$	equivalent ckt at $t = \infty$ steady state

uncharged capacitor = SC
charged capacitor = voltage source

8. For the n/w shown the switch is closed at $t = 0$. find initial condition $i(0^+)$ & $\frac{di(0^+)}{dt}$



Solution

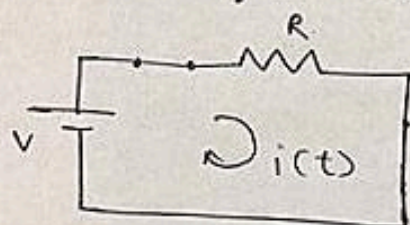
→ at $t = 0^-$ switch is open

$$i(0^-) = 0$$

$$V_c(0^-) = 0$$

$$V(0) = 0$$

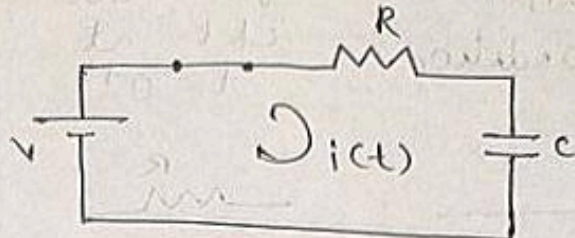
→ at $t = 0^+$ switch closed



$$i(t) = \frac{V}{R}$$

$$i(0^+) = \frac{V}{R} \text{ A}$$

→ at $t > 0$



$$V = V_R + V_C$$

$$V = R i(t) + \frac{1}{C} \int i(t) dt$$

differentiate wrt t

$$0 = R \frac{di(t)}{dt} + \frac{1}{C} i(t)$$

$$R \frac{di(t)}{dt} = -\frac{1}{C} i(t)$$

at $t = 0^+$

$$R \frac{di(0^+)}{dt} = -\frac{1}{C} i(0^+)$$

$$R \frac{di(0^+)}{dt} = -\frac{1}{C} \left[\frac{V}{R} \right]$$

$$\frac{di(0^+)}{dt} = -\frac{1}{R} \frac{V}{C}$$

$$\boxed{\frac{di(0^+)}{dt} = -\frac{V}{R^2 C} \text{ A/s}}$$



at $t = 0^+$

$$i(0^+) = 0$$

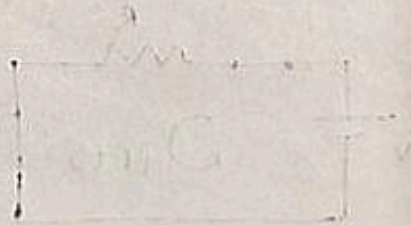
$$V(0^+) = V$$

$$V(0^+) = V$$

at $t = 0^+$

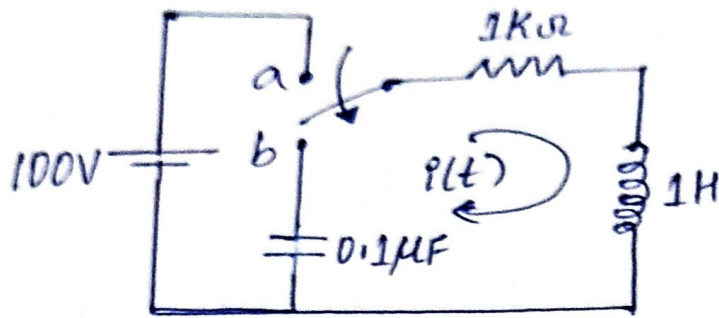
$$\frac{V}{R} i(0^+)$$

$$\frac{V}{R} i(0^+)$$

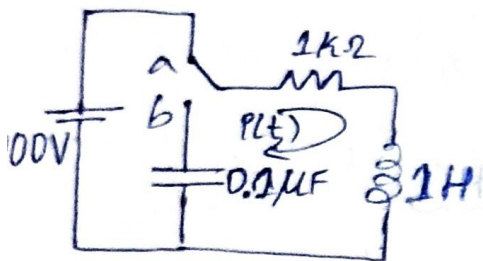


MODULE-04 [Unit → 02]

① In the Network shown, the switch is changed from the position 'a' to 'b' at $t=0$. Solve for i , $\frac{di}{dt}$ & $\frac{d^2i}{dt^2}$ at $t=0^+$



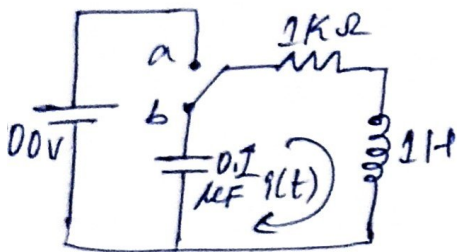
⇒ Step 1:- When initial condition $t=0^-$
Switch is at 'a' & inductor behaves as short circuit



$$i(0^-) = \frac{100V}{1k\Omega} = 0.1A$$

$$V_C(0^-) = 0$$

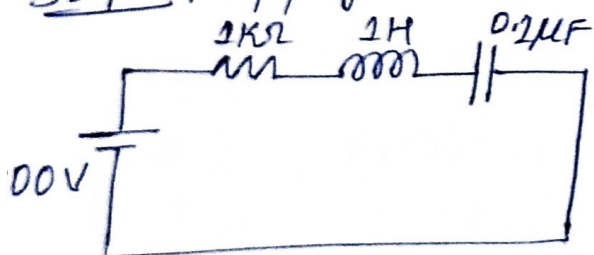
Step 2:- When initial condition $t=0^+$
Switch is at 'b'



$$i(0^+) = i(0^-) = 0.1A$$

$$V_C(0^+) = 0$$

Step 3:- Apply KVL for RLC circuit



$$Ri + L \frac{di}{dt} + V_C = 0$$

$$1000 + 1 \frac{di}{dt} + 0.1 = 0$$

at $t=0^+$ $\frac{di(0^+)}{dt} = -100A/s$

Step 4:- differentiate again

$$L \frac{d^2 i}{dt^2} + R \frac{di}{dt} + \frac{1}{C} i = 0$$

$$1 \frac{d^2 i}{dt^2} + 1000(-100) + \frac{1}{0.1 \times 10^{-6}} (0.1) = 0$$

$$\frac{d^2 i}{dt^2} - 10^5 + 10^6 = 0$$

$$\frac{d^2 i}{dt^2} (0^+) = -9 \times 10^5$$

$$\boxed{\frac{d^2 i}{dt^2} (0^+) = -900000 \text{ A/s}^2}$$

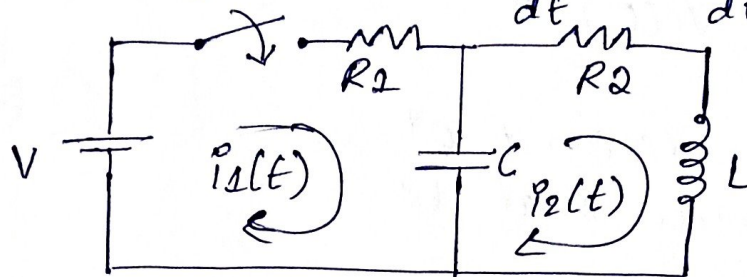
$$\underline{01} \quad \boxed{-9 \times 10^5 \text{ A/s}^2}$$

$$C = 0.1 \times 10^{-6} (0.1 \mu\text{F})$$

$$i(0^+) = 0.1$$

$$\frac{di}{dt} (0^+) = -100$$

② In the Network shown, Assuming all initial conditions as zero, Find $i_1(0^+)$, $i_2(0^+)$, $\frac{di_1}{dt}(0^+)$, $\frac{di_2}{dt}(0^+)$, $\frac{d^2 i_1}{dt^2}(0^+)$, $\frac{d^2 i_2}{dt^2}(0^+)$.



Note:-

$\Rightarrow$ Two mesh currents $\rightarrow i_1(t) \rightarrow$ left loop
 $i_2(t) \rightarrow$ Right loop

- capacitor is common
- Switch closes at $t=0$

Step 1:- at $t=0^-$

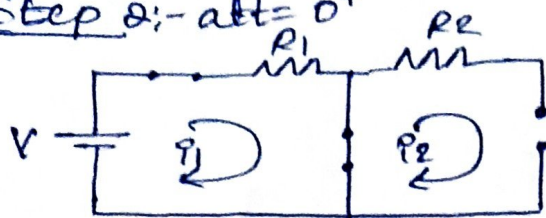
The circuit is open and No supply to loops

$$i_1(0^-) = 0$$

$$i_2(0^-) = 0$$

$$V_C(0^-) = 0$$

Step 2:- at $t = 0^+$



Capacitor is shorted
 $V_C = 0$

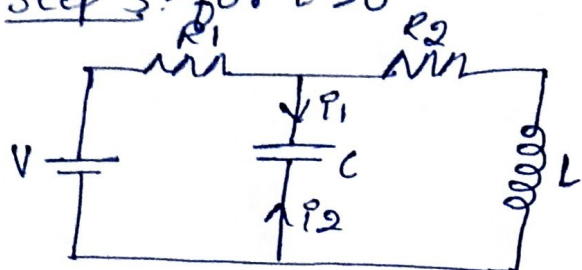
$$i_1(0^+) = \frac{V}{R_1}$$

Inductor is open circuit, so no current flows

$$i_2(0^+) = 0$$

$$V_C(0^+) = 0$$

Step 3:- for $t > 0$



$$V = R_1 i_1 + V_C$$

$$0 = R_1 \frac{di_1}{dt} + \frac{dV_C}{dt}$$

$$\frac{di_1}{dt} = -\frac{1}{R_1} \frac{dV_C}{dt}$$

$$V_C = \frac{1}{C} \int (i_1 - i_2) dt$$

$$\frac{dV_C}{dt} = \frac{1}{C} (i_1 - i_2)$$

$$\text{at } t = 0^+, \frac{dV_C}{dt} = \frac{1}{C} [i_1(0^+) - i_2(0^+)]$$

$$\text{but, } i_2(0^+) = 0 \text{ \& } i_1(0^+) = \frac{V}{R_1}$$

$$\therefore \frac{dV_C}{dt} = \frac{1}{C} \cdot \frac{V}{R_1}$$

$$\frac{dV_C}{dt} = \frac{V}{R_1 C}$$

$$\frac{di_1(0^+)}{dt} = -\frac{V}{R_1^2 C}$$

Step 4:- Differentiate again

$$L \frac{di_2}{dt} = V_C$$

$$\therefore V_C = 0$$

$$\frac{di_2(0^+)}{dt} = 0$$

Step 5:- for i_2 , $i_1 - i_2 = C \frac{dV_C}{dt}$

$$\frac{di_1}{dt} - \frac{di_2}{dt} = C \frac{d^2 V_C}{dt^2}$$

$$-\frac{V}{R_1^2 C} = C \frac{d^2 V_C}{dt^2}$$

$$\frac{d^2 i_1}{dt^2} = -\frac{V}{R_1^2 C^2}$$

$$\frac{d^2 i_1}{dt^2} = -\frac{1}{R_1} \cdot \frac{d^2 V_C}{dt^2}$$

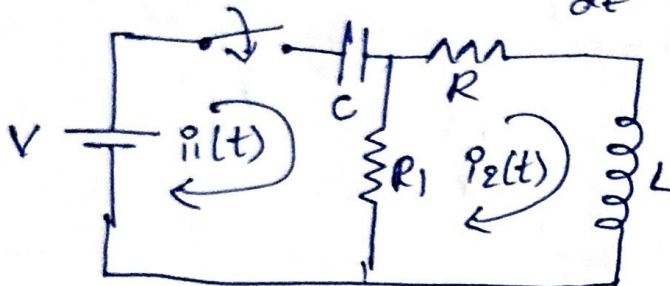
$$\frac{d^2 i_1}{dt^2} = \frac{V}{R_1^2 C^2}$$

$$\boxed{\frac{d^2 i_1}{dt^2} = \frac{V}{R_1^2 C^2}}$$

for i_2 , $L = \frac{d^2 i_2}{dt^2} = \frac{dV_C}{dt}$

$$\boxed{\frac{d^2 i_2}{dt^2} = \frac{V}{LR_1 C}}$$

③ In the N/w shown, Assuming all initial conditions as zero, Find $i_1(0^+)$, $i_2(0^+)$, $\frac{di_1}{dt}(0^+)$, $\frac{di_2}{dt}(0^+)$, $\frac{d^2 i_1}{dt^2}(0^+)$, $\frac{d^2 i_2}{dt^2}(0^+)$



⇒ Step 1:- at $t=0^-$

all initial conditions are zero

$$V_C(0^-) = 0$$

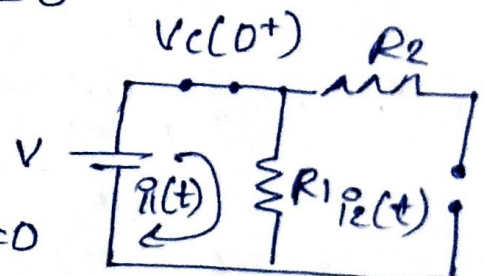
$$i_1(0^-) = 0$$

$$i_2(0^-) = 0$$

Step 2:- at $t=0^+$

Capacitor acts as short circuit &
Inductor acts as open circuit.

$$\boxed{i_1(0^+) = V/R_1} \quad \boxed{i_2(0^+) = 0} \quad V_C(0^+) = 0$$

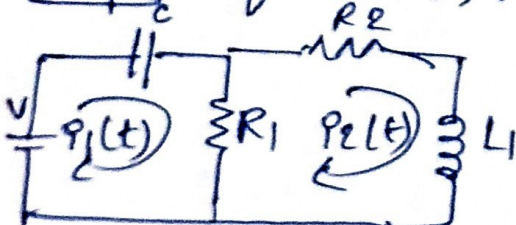


Step 3:- for $t > 0$, Apply KVL $\Rightarrow V - \frac{1}{C} \int i_1 dt - R_1(i_1 - i_2) = 0 \rightarrow (1)$

$$-R_1(i_2 - i_1) - R_2 i_2 - L \frac{di_2}{dt} = 0 \rightarrow (2)$$

from (2), $\frac{di_2}{dt} = \frac{1}{L} [R_1 i_1 - (R_1 + R_2) i_2]$

$$\frac{di_2}{dt}(0^+) = \frac{1}{L} [R_1 i_1(0^+) - (R_1 + R_2) i_2(0^+)] \rightarrow (3)$$



$$\frac{di_2}{dt}(0^+) = \frac{1}{L} \left[R_2 \frac{V}{R_1} - (R_1 + R_2) 0 \right] = \frac{V}{L}$$

differentiating eqⁿ (2)

$$0 = \frac{i_2}{C} - R_1 \frac{di_1}{dt} + R_2 \frac{di_2}{dt} = 0$$

$$\left[\frac{di_1}{dt} = \frac{di_2}{dt} = \frac{i_1}{R_1 C} \right] \rightarrow (4)$$

$$\text{at } t=0^+ \quad \frac{di_1(0^+)}{dt} = \frac{di_2(0^+)}{dt} = \frac{i_1(0^+)}{R_1 C} = \frac{V}{L} - \frac{V}{R_1^2 C} //$$

Step 4:- differentiate again (3)

$$\frac{d^2 i_2}{dt^2} = \frac{1}{L} \left[R_1 \frac{di_1}{dt} - (R_1 + R_2) \frac{di_2}{dt} \right]$$

$$\text{at } t=0^+ \quad \left[\frac{d^2 i_2(0^+)}{dt^2} = -V \left[\frac{1}{R_1 LC} + \frac{R_2}{L^2} \right] \right]$$

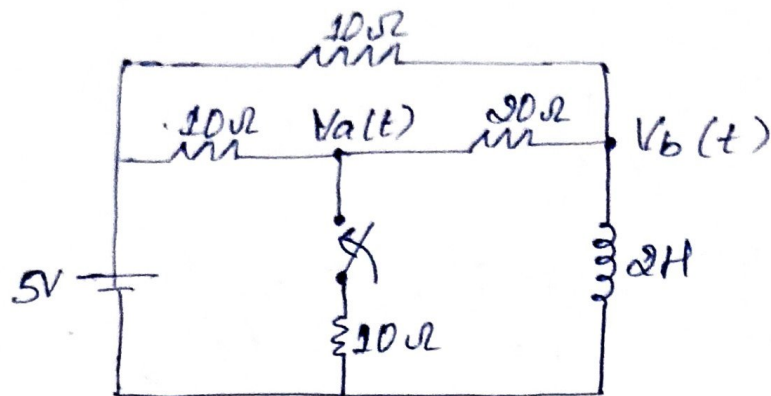
Step 5:- differentiate eqⁿ (4)

$$\frac{d^2 i_1}{dt^2} = \frac{d^2 i_2}{dt^2} - \frac{1}{R_1 C} \frac{di_1}{dt}$$

$$\text{at } t=0^+, \quad \frac{d^2 i_1(0^+)}{dt^2} = \frac{d^2 i_2(0^+)}{dt^2} - \frac{1}{R_1 C} \frac{di_1(0^+)}{dt}$$

$$\left[\frac{d^2 i_1(0^+)}{dt^2} = -\frac{V}{R_1 LC} - \frac{VR_2}{L^2} - \frac{1}{R_1 C} \left[\frac{V}{R_1^2 C} - \frac{2V}{R_1 LC} \right] \right]$$

④ In the network shown, a steady state is reached with the switch open. At $t=0$, the switch is closed. For the element values given, determine the value of $V_a(0^-)$, $V_b(0^-)$, $V_a(0^+)$ and $V_b(0^+)$.



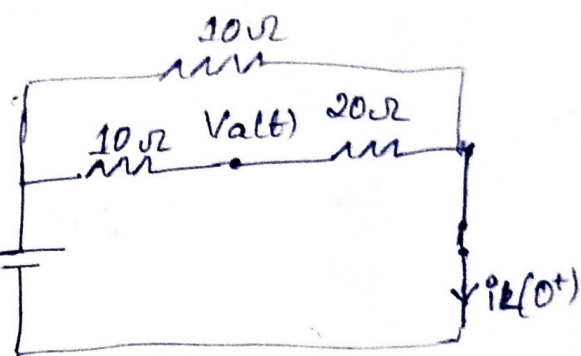
⇒ Step 1 :- at $t=0^-$

the network attains steady state condition & inductor acts short ckt & 5V

$$i_L(0^-) = \frac{5}{(10 \parallel 30)} = \frac{5}{7.5} = \frac{2}{3} \text{ A}$$

$$V_b(0^-) = 0$$

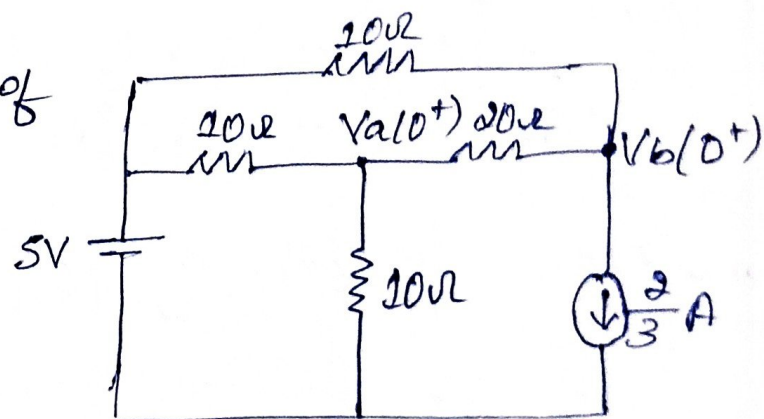
$$V_a(0^-) = 5 \times \frac{20}{30} = 3.33 \text{ V}$$



Step 2 :- at $t=0^+$

Inductor acts a current source of $\frac{2}{3} \text{ A}$.

$$i_L(0^+) = \frac{2}{3} \text{ A}$$



Step 3 :- Apply KCL at $t=0^+$

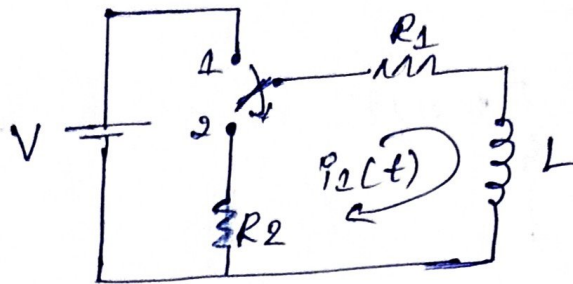
$$\frac{V_a(0^+) - 5}{10} + \frac{V_a(0^+)}{10} + \frac{V_a(0^+) - V_b(0^+)}{20} = 0 \rightarrow (1)$$

$$\& \frac{V_b(0^+) - V_a(0^+)}{20} + \frac{V_b(0^+) - 5}{10} + \frac{2}{3} = 0 \rightarrow (2)$$

Solving eqn (1) & (2) we get,

$$\begin{aligned} V_a(0^+) &= 1.9V \\ V_b(0^+) &= -0.477V \end{aligned}$$

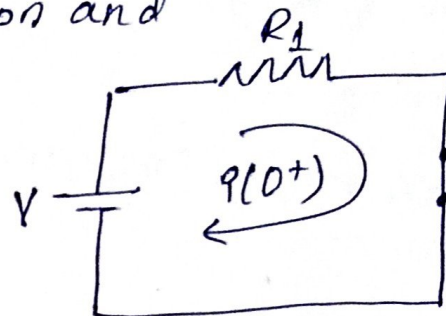
⑤ In the network shown, the switch is initially at the position 1. On the steady state having reached, the switch is changed to the position 2. Find the current $i(t)$



Step 1 :- at $t=0^-$

It attains steady state condition and Inductor acts as short ckt

$$i(0^-) = \frac{V}{R_1}$$

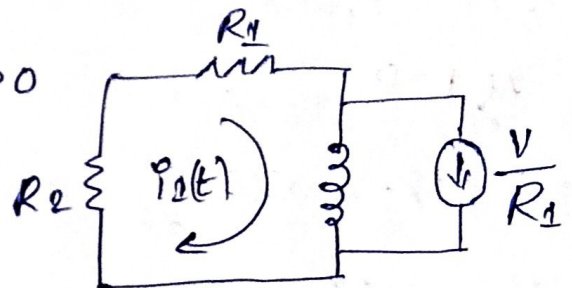


Step 2 :- $i(0^+) = \frac{V}{R_1}$

Step 3 :- for $t > 0$, Apply KVL for $t > 0$

$$-R_2 i - L \frac{di}{dt} = 0$$

$$\frac{di}{dt} + \frac{(R_1 + R_2)}{L} i = 0$$



comparing with differential eqn $\frac{di}{dt} + Pi = 0$

$$P = \frac{R_1 + R_2}{L}$$

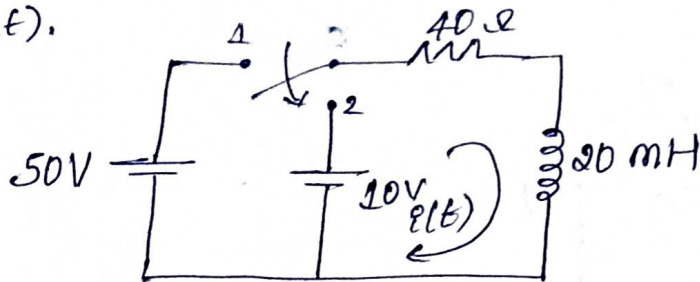
The soln. to differential eqn. is $i(t) = Ke^{-Pt}$
 $i(t) = Ke^{-\left(\frac{R_1 + R_2}{L}\right)t}$

~~$i(t) =$~~
at $t=0$, $i(0) = \frac{V}{R_1}$

$$\frac{V}{R_1} = Ke^0 = K$$

$$i(t) = \frac{V}{R_1} e^{(-\frac{R_1+R_2}{L})t} \quad \text{for } t > 0.$$

⑥ For the network shown, it is under steady state with switch at position 1. At $t=0$, switch is moved to position 2. Find $i(t)$.



⇒ Step 1: at $t=0^-$

network attains steady state condition & Inductor acts as short ckt

$$i(0^-) = \frac{50}{40} = 1.25A$$

Step 2: at $t=0^+$
Since current through inductor cannot change instantaneously

$$i(0^+) = 1.25A$$

Step 3: At $t > 0$
Apply KVL eqn

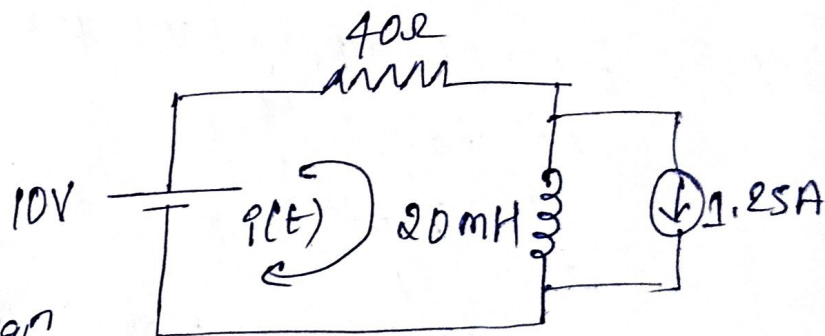
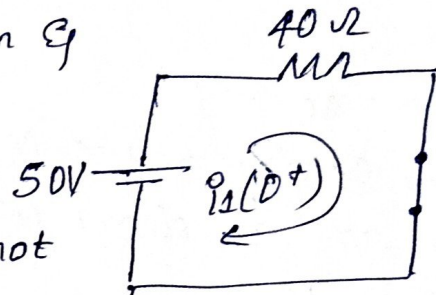
$$10 - 40i - 20 \times 10^{-3} \frac{di}{dt} = 0$$

$$\frac{di}{dt} + 2000i = 500$$

comparing with differential eqn

$$\frac{di}{dt} + pi = Q$$

$$p = 2000, \quad Q = 500$$



$$\begin{aligned}
 i(t) &= e^{-Pt} \int Q e^{Pt} dt + K e^{-Pt} \\
 &= e^{-2000t} \int 500 e^{2000t} + K e^{-2000t} \\
 &= \frac{500}{2000} + K e^{-2000t} \\
 i(t) &= 0.25 + K e^{-2000t}
 \end{aligned}$$

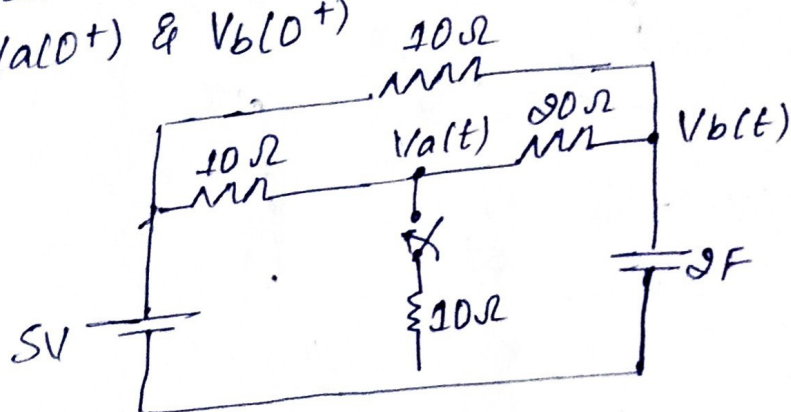
at $t=0$,
 $i(0) = 1.25A$

$$1.25 = 0.25 + K$$

$$K = 1$$

$$i(t) = 0.25 + e^{-2000t}$$

⑦ In the network shown, a steady state is reached with the switch open. At $t=0$, the switch is closed. For the element values given, determine the value of $V_a(0^-)$, $V_b(0^-)$, $V_a(0^+)$ & $V_b(0^+)$



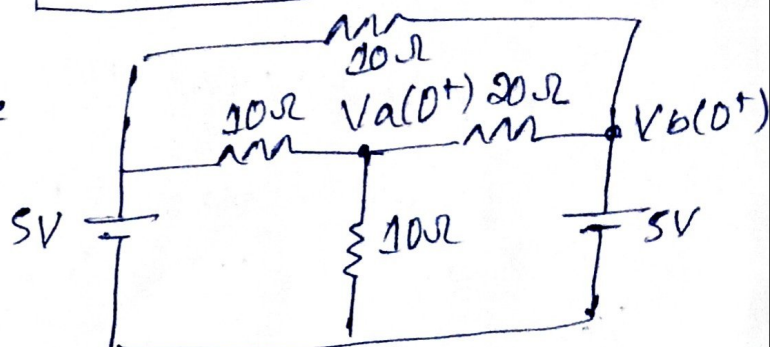
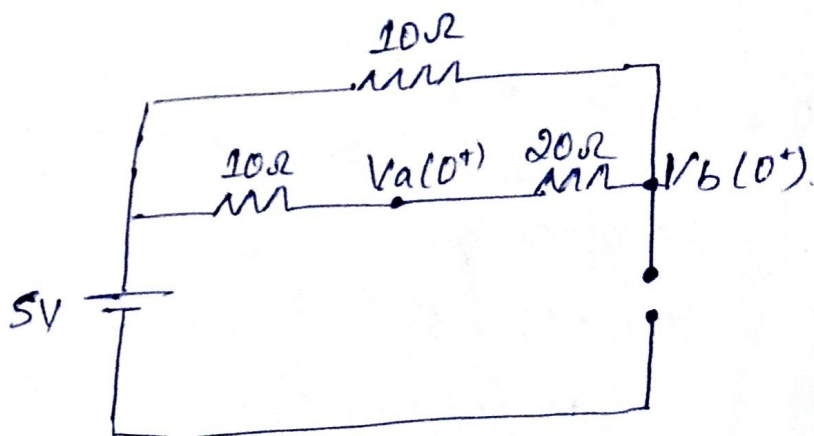
⇒ Step 1:- at $t=0^-$,
 capacitor acts as open ckt,

$$V_a(0^-) = 5V$$

$$V_b(0^-) = 5V$$

Step 2:- at $t=0^+$,
 capacitor acts as voltage source
 of 5V

$$V_b(0^+) = 5V$$



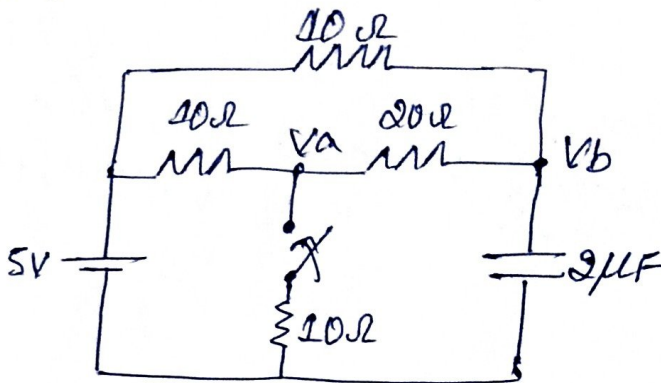
Step 3: Apply KCL at eqn $t=0^+$

$$\frac{V_a(0^+) - 5}{10} + \frac{V_a(0^+)}{10} + \frac{V_a(0^+) - 5}{20} = 0$$

$$0.25 V_a(0^+) = 0.75$$

$$\boxed{V_a(0^+) = 3V}$$

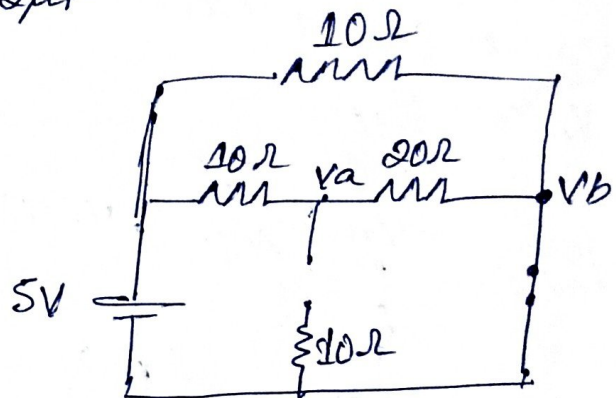
⑧ In the circuit shown, the steady state is reached with switch is open. At $t=0$, switch K is closed. Find the value of $V_a(0^-)$ and $V_a(0^+)$.



⇒ at Step 1 :- at $t=0^-$.
Capacitor acts as open ckt

$$V_b(0^-) = 5V$$

$$V_a(0^-) = 5V$$



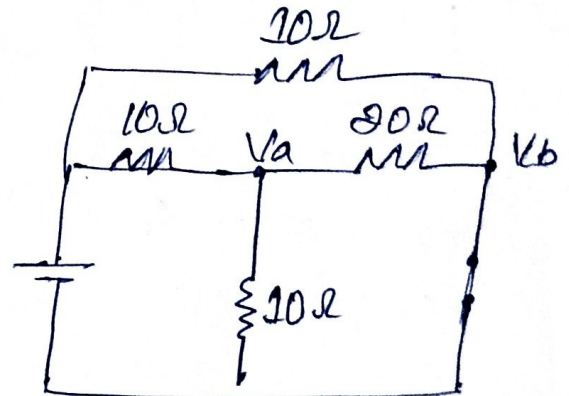
Step 2 :- at $t=0^+$
Capacitor voltage cannot change instantaneously.

$$V_c(0^+) = V_c(0^-) = 5V$$

$$V_b(0^+) = 5V$$

Step 3 :- Apply KCL at node $V_a(0^+)$

$$\frac{V_a - 5}{10} + \frac{V_a}{10} + \frac{V_a - 5}{20} = 0$$



$$2(V_a - 5) + 2V_a + (V_a - 5) = 0$$

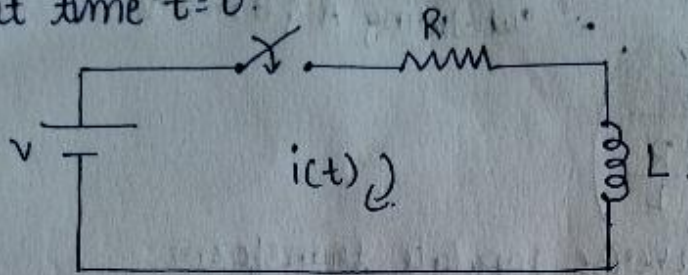
$$2V_a - 10 + 2V_a + V_a - 5 = 0$$

$$5V_a - 15 = 0$$

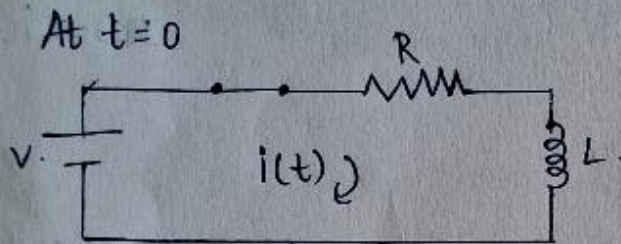
$$\boxed{V_a = 3V}$$

Module 4 : Unit 3.

- 1) Find $i(t)$ for the series RL circuit given, the switch is closed at time $t=0$.



Solution:-



Apply KVL

$$V = i(t)R + L \frac{di(t)}{dt}$$

Apply Laplace Transform

$$\frac{V}{s} = I(s)R + L \{ sI(s) - i(0) \}$$

$$\frac{V}{s} = I(s)R + LsI(s)$$

$$\frac{V}{s} = I(s) [R + Ls]$$

$$I(s) = \frac{V/s}{(R + Ls)} \Rightarrow \frac{V/s}{L(\frac{R}{L} + s)}$$

Common

$$I(s) = \frac{A}{s} + \frac{B}{(s + \frac{R}{L})} \rightarrow \text{①}$$

To find A.

$$A = \lim_{s \rightarrow 0} \frac{\frac{V}{L}}{(s + \frac{R}{L})} = \frac{\frac{V}{L}}{\frac{R}{L}} = \frac{V}{R}$$

$$B = \lim_{s \rightarrow -\frac{R}{L}} \frac{\frac{V}{L}}{s} \Rightarrow \frac{\frac{V}{L}}{-\frac{R}{L}} \Rightarrow -\frac{V}{R}$$

$$I(s) = \frac{\frac{V}{R}}{s} + \left(\frac{-\frac{V}{R}}{s + \frac{R}{L}} \right) \quad \rightarrow \text{substituting A \& B in eq ①}$$

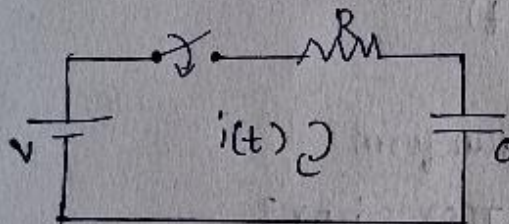
Apply Inverse Laplace transform

$$i(t) = \frac{V}{R} - \frac{V}{R} e^{-\frac{R}{L}t}$$

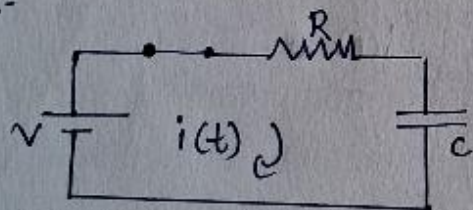
$$I(s) \Rightarrow i(t)$$

$$i(t) = \frac{V}{R} (1 - e^{-\frac{R}{L}t})$$

2). Find $i(t)$ for series RC circuit where switch is closed at $t=0$



Sol:-



Apply KVL

$$V = i(t)R + \frac{1}{C} \int_0^t i(t) dt + V(0)$$

Apply Laplace transform

$$\frac{V}{s} = I(s)R + \frac{1}{Cs} I(s) + \frac{V(0)}{s}$$

initial condit
is 0 so
 $V(0) = 0$

$$\frac{V}{s} = I(s) \left(R + \frac{1}{Cs} \right)$$

$$I(s) = \frac{\frac{V}{s}}{R + \frac{1}{Cs}} \Rightarrow \frac{\frac{V}{s}}{\left(\frac{RCs + 1}{s} \right)}$$

$$I(s) = \frac{V_c}{1 + RCs} = \frac{V_c}{RC \left(\frac{1}{RC} + s \right)}$$

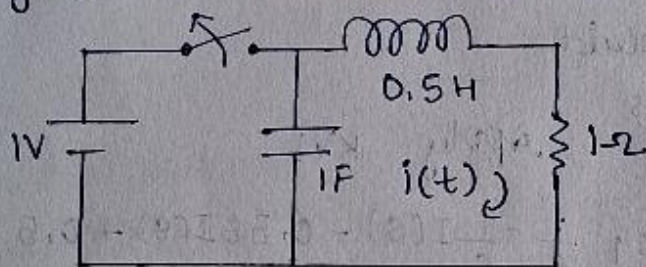
$$= \frac{V/R}{\left(s + \frac{1}{RC} \right)}$$

Apply Inverse Laplace transform

$$i(t) = \frac{V}{R} e^{-\frac{t}{RC}}$$

$$i(t) = \frac{V}{R} e^{-t/RC}$$

- 3) In the Network shown in figure, the switch is opened at $t=0$, steady state condition is achieved before $t=0$. Find $i(t)$



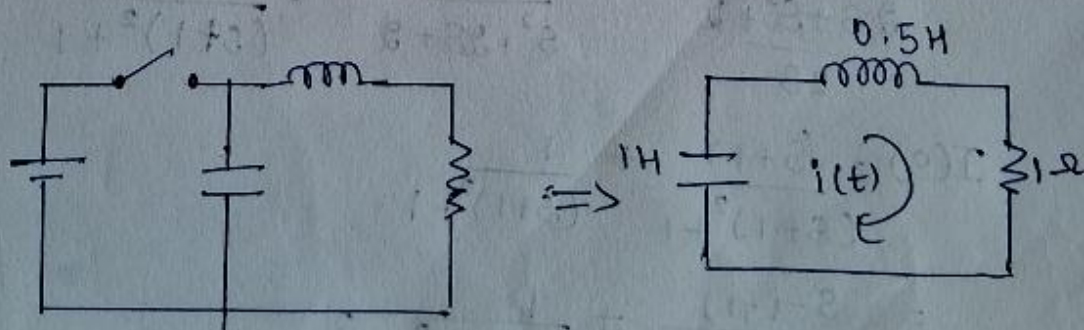
Sol: At $t=0^-$, switch is closed and steady state condition is achieved, Hence, the capacitor acts as an open circuit & inductor behaves as a short circuit

$$V_c(0^-) = 1V \quad i(0^-) = 1A$$

Since current through the inductor & voltage across the capacitor cannot change instantaneously.

$$V_c(0^+) = 1V \quad i(0^+) = 1A$$

for $t > 0$



Apply KVL

$$V_0 + V_L + V_R = 0$$

$$\frac{1}{C} \int_0^t i dt + V(0) + L \frac{d}{dt} i(t) + i(t)R = 0$$

$$1 \int_0^t i(t) dt + V(0) + 0.5 \frac{d}{dt} i(t) + i(t) = 0$$

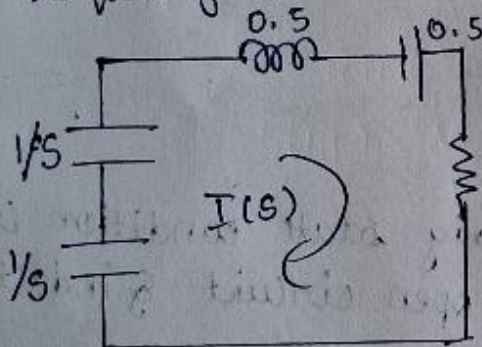
Apply Laplace transform

$$1 \left[\frac{I(s)}{s} + \frac{V(0)}{s} \right] + 0.5 [sI(s) - i(0)] + I(s) = 0$$

$$V(0^+) = 1V \quad i(0^+) = 1A$$

$$\frac{I(s)}{s} + \frac{1}{s} + 0.5sI(s) - 0.5 + I(s) = 0$$

Frequency domain circuit



Apply KVL

$$\frac{1}{s} - \frac{1}{s} I(s) - 0.5sI(s) + 0.5 - I(s) = 0$$

$$\frac{1}{s} I(s) + 0.5s I(s) + I(s) = \frac{1}{s} + 0.5$$

$$I(s) \left[\frac{1}{s} + 0.5s + 1 \right] = \frac{1}{s} + 0.5$$

$$I(s) = \frac{0.5 + \frac{1}{s}}{\left(1 + 0.5s + \frac{1}{s}\right)} = \frac{\frac{1}{2} + \frac{1}{s}}{\left(1 + \frac{1}{2}s + \frac{1}{s}\right)}$$

$$I(s) = \frac{S+2}{2S} \cdot \frac{2S}{2S^2+S^2+2}$$

$$= \frac{S+2}{S^2+2S+2} = \frac{(S+1)+1}{(S^2+1)^2+1}$$

$$I(s) = \frac{(S+1)}{(S+1)^2+1} + \frac{1}{(S+1)^2+1}$$

$$= \frac{S-(-1)}{(S+1)^2+1^2} + \frac{1}{(S-(-1))^2+1^2}$$

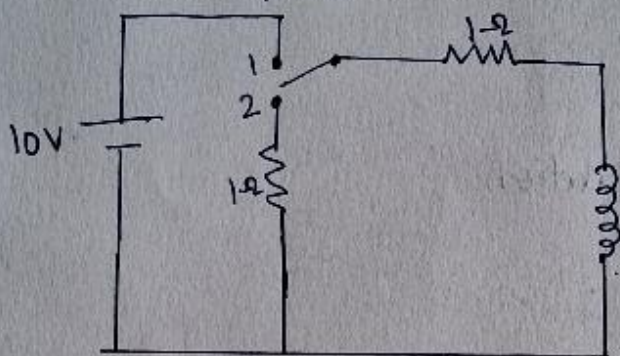
Compare with standard form $a = -1$ $w = 1$

Apply Inverse Laplace transform

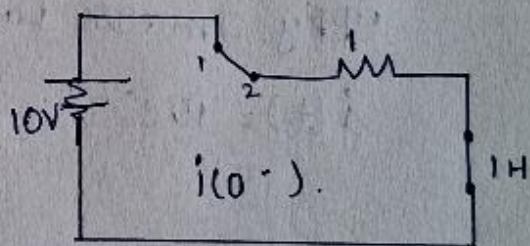
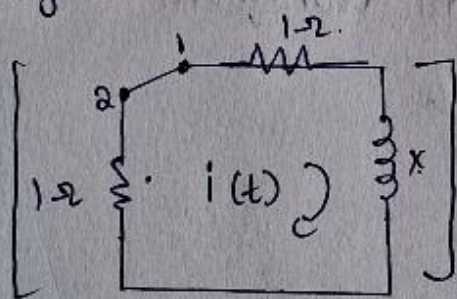
$$i(t) = e^{-t} \cos t + e^{-t} \sin t$$

$$i(t) = e^{-t} (\cos t + \sin t)$$

- 4) In the Network shown, the switch is moved from the position 1 to 2 at $t = 0$, steady state condition having been established in the position 1. Determine $i(t)$ for $t > 0$

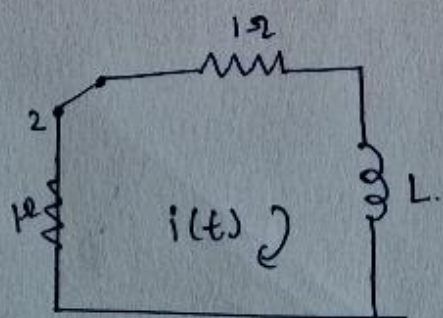


Sol:- At $t = 0^-$, the network is as shown. The circuit has obtained steady state condition, hence inductor acts as short circuit



$i(0^-) = \frac{10}{1} = 10A$. Since current through the inductor cannot change instantaneously $i(0^+) = 10A$

for $t > 0$.



Apply KVL

$$1i(t) + 1 \cdot i(t) + L \frac{di(t)}{dt} = 0$$

Apply Laplace transform

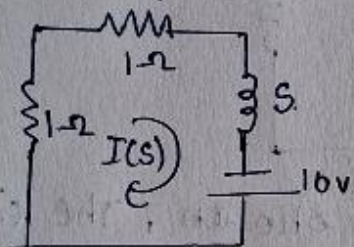
$$I(s) + I(s) + 1 [sI(s) - i(0)] = 0$$

$$I(s) + I(s) + sI(s) - 10 = 0$$

$$I(s) \cdot [s+2] - 10 = 0$$

$$I(s) = \frac{10}{s+2}$$

Frequency domain representation.



Apply Inverse Laplace transform

$$i(t) = 10e^{-2t}, \quad t > 0$$